Online state and parameter estimation for Bloch–Torrey dynamics

A total-sensitivity adaptive observer with cumulative output derivatives

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Outline

1. MRI, Bloch–Torrey and raw MRAS
2. Output jets and total sensitivity
3. Convergence and Bloch–Torrey numerics

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3 Convergence and Bloch–Torrey numerics

MRI can estimate coefficients of the dynamics

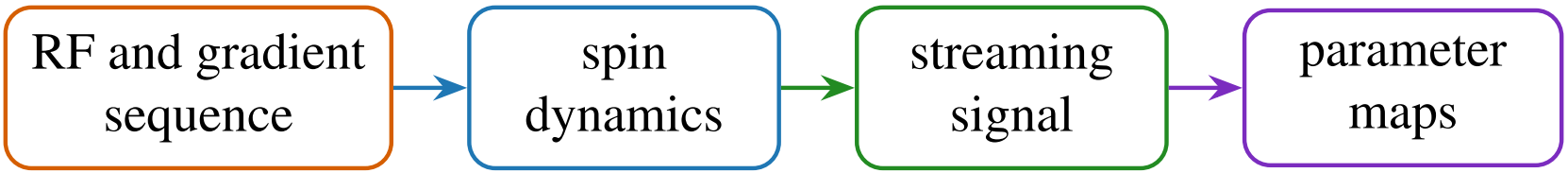
Conventional reconstruction

The acquisition produces an image whose contrast depends on the sequence and on several tissue effects.

Quantitative reconstruction

The dynamic signal is used to infer

$$T_1,\ T_2,\ \Delta f,\ M_{\mathrm{eq}}.$$



Online objective

Update the state and the tissue parameters while the experiment is running.

Online inverse problem

The Bloch–Torrey dynamics are

$$\partial_t M^\star - D\Delta M^\star = \mathcal{C}(t,x;\omega)M^\star - \operatorname{diag}(r_2, r_2, r_1)M^\star + (0,0,\rho)^\top.$$

Unknown coefficients

$$\theta^\star = (r_2, r_1, \omega, \rho),$$

$$r_j = T_j^{-1}, \qquad M_{\mathrm{eq}} = \rho / r_1.$$

Each coefficient may be a spatial field.

Partial observation

$$y(t) = RM^\star(t) = CM_\perp^\star(t),$$

$$M_\perp = M_x + \mathrm{i}M_y.$$

The receiver measures transverse magnetisation. It does not measure M_z directly.

Goal

Recover $M(t)$ and θ causally from the stream $y(t)$.

Starting point: raw-output MRAS

Write the Bloch–Torrey equation in the affine form

$$\partial_t M = F_0(t, M) - B(M)\theta, \qquad y = RM.$$

With the measured residual $r = R\widehat{M} - y$, raw-output MRAS uses

$$\partial_t \widehat{M} = F_0(t, \widehat{M}) - B(\widehat{M})\widehat{\theta} - \mu R^* r, \qquad \text{state observer,}$$

$$\partial_t \widehat{\theta} = \gamma B(\widehat{M})^\sharp R^* r, \qquad \text{parameter update.}$$

Attractive features

- online and low-memory,
- one residual for both equations,
- an energy cancellation in affine settings.

Regressor used by the update

$$RB(\widehat{M}(t)).$$

This describes the immediate parameter forcing visible through the receiver.

Raw MRAS cannot update longitudinal parameters

In transverse/longitudinal notation,

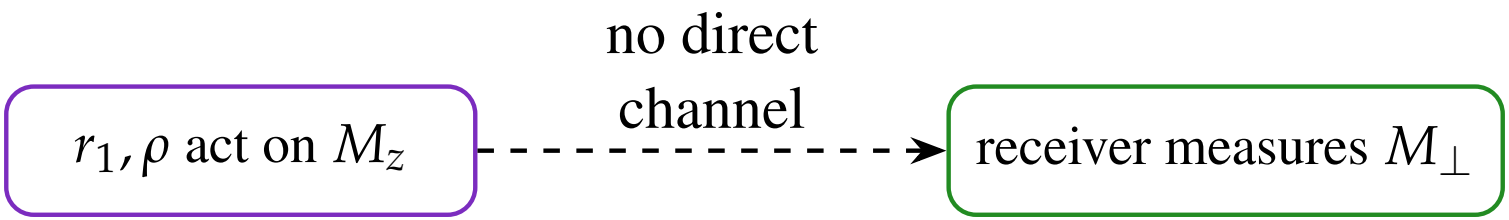
$$B(M)\theta = \big((r_2 + \mathrm{i}\omega)M_\perp,\ r_1 M_z - \rho\big).$$

Invariant longitudinal kernel

For every state, every time, and every excitation,

$$\{(0, \xi_1, 0, \xi_\rho)\} \subset \ker\big(RB(M)\big).$$

Hence the raw-MRAS updates of r_1 and ρ vanish identically.



Scope of the result. RF rotation can still encode r_1 and ρ in later transverse data. The obstruction concerns the instantaneous MRAS regressor.

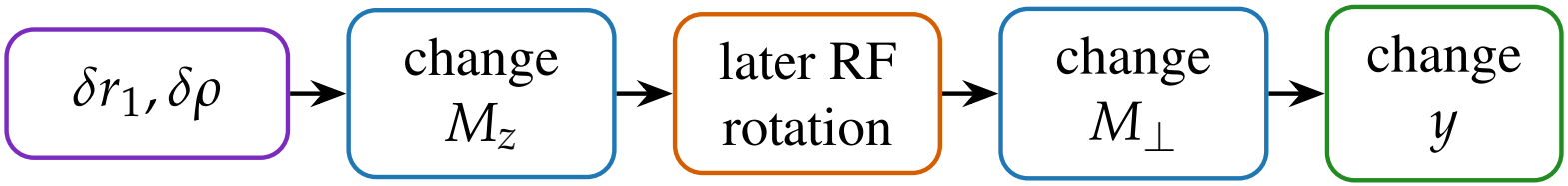
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Longitudinal information appears after propagation



1. **Output derivatives** compress a short-time response into a local feature: $y, \dot{y}, \ddot{y}, \ldots$
2. **Trajectory sensitivity** propagates how the observer changes when the parameter changes.

Strategy

Use both descriptions of the dynamics, then compare their respective contributions.

Output derivatives expose the local response

Define the cumulative output jet

$$\Psi_m(t, M, \theta) = \left(y, \dot{y}, \ldots, y^{(m)}\right).$$

First derivative

$$D_\theta \Phi_1 \, \xi = -C\left[M_\perp (\xi_2 + \mathrm{i}\xi_\omega)\right].$$

It sees the transverse pair (r_2, ω) directly.

Second derivative

$$D_\theta \Phi_2(0, \xi_1, 0, \xi_\rho) = C\left[h(\xi_\rho - M_z \xi_1)\right].$$

During an RF pulse, it also sees the longitudinal pair (r_1, ρ).

Interpretation

The receiver observes how M_z changes the acceleration of the transverse signal.

First attempt: direct jet sensitivity

Let

$$\rho_m = \Psi_m(t, \widehat{M}, \widehat{\theta}) - \mathbf{y}_m, \qquad K_m = D_\theta \Psi_m(t, \widehat{M}, \widehat{\theta}) \quad (\widehat{M} \text{ fixed}).$$

A natural update is

$$\partial_t \widehat{\theta} = -\beta K_m^* \rho_m.$$

What this captures

The order-two Bloch blocks distinguish the four parameter types over a suitable time window.

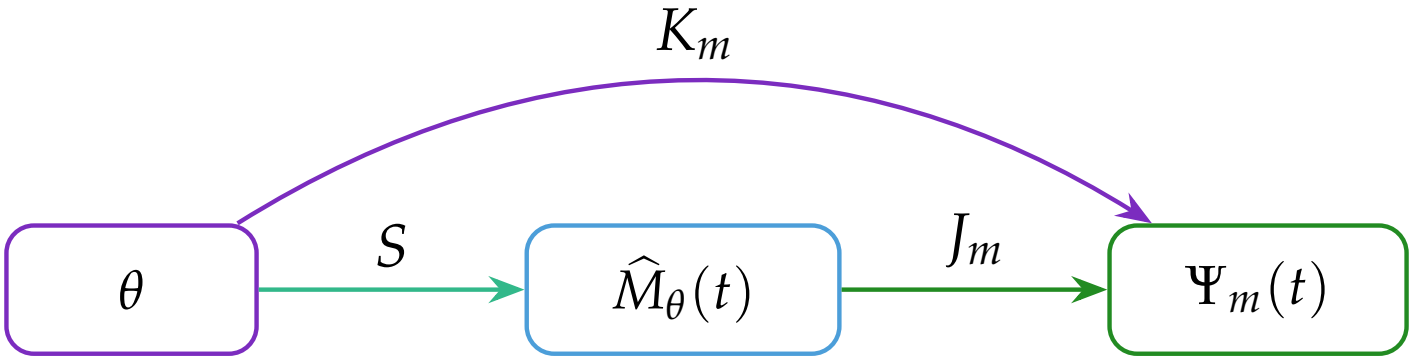
Yet K_m freezes the state. Changing $\widehat{\theta}$ also changes the observer trajectory, and this indirect effect is missing.

The direct update can therefore point in the wrong direction for the coupled estimator.

Total derivative of the predicted jet

Consider the frozen-parameter family of observer trajectories $\widehat{M}_\theta(t)$ and define

$$S(t) = \partial_\theta \widehat{M}_\theta(t), \qquad J_m = D_M \Psi_m.$$



$$\frac{d}{d\theta}\Psi_m(t, \widehat{M}_\theta(t), \theta) = \underbrace{K_m}_{\text{direct}} + \underbrace{J_m S}_{\text{through the trajectory}} =: H_m.$$

Role of the sensitivity

H_m is the total frozen-parameter derivative of the predicted jet. Stability still requires observer decay, excitation, and a coupling bound.

Cumulative total-sensitivity estimator

$$\partial_t \widehat{M} = F(t, \widehat{M}, \widehat{\theta}) - \mu R^*(R\widehat{M} - y),$$
$$\partial_t \widehat{S} = \left[D_M F(t, \widehat{M}, \widehat{\theta}) - \mu R^* R\right]\widehat{S} + D_\theta F(t, \widehat{M}, \widehat{\theta}),$$
$$\partial_t \widehat{\theta} = -\beta \widehat{H}_m^* \rho_m, \qquad \widehat{H}_m = K_m + J_m \widehat{S}.$$

Order zero already contains memory

Since

$$K_0 = 0, \qquad J_0 = R, \qquad H_0 = R\widehat{S},$$

the residual contains only raw output, while the regressor contains the propagated history of parameter effects.

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Three ingredients for stable estimation

State decay

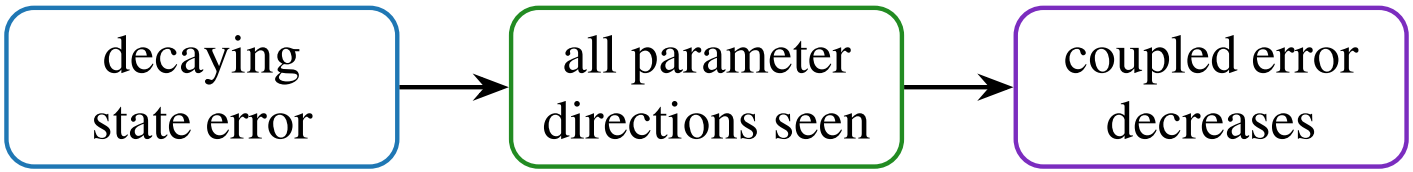
Positive r_1, r_2 damp state errors, while nudging corrects the measured transverse component.

Sequence excitation

RF pulses repeatedly carry hidden M_z information into the measured $M_\perp$ signal.

Time-scale balance

The state and sensitivity must track faster than the parameters are updated.



What we can verify for Bloch–Torrey

Analytically: relaxation damps state perturbations

$$\frac{1}{2}\frac{d}{dt}\|v\|^2 \leq -r_{\min}\|v\|^2, \qquad r_1, r_2 \geq r_{\min} > 0.$$

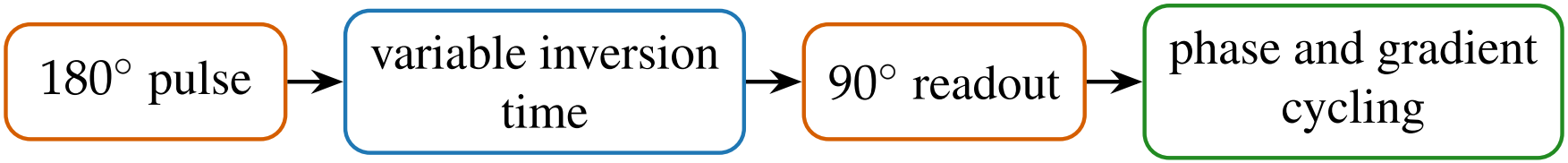
Numerically: the repeated sequence supplies information

In the current discretisation, the sensitivity Gramian sees all twelve coefficients and the coupled linearisation is stable.

Still open at PDE level

We still need a continuum excitation estimate and a bound showing that the observer decay dominates the remaining couplings.

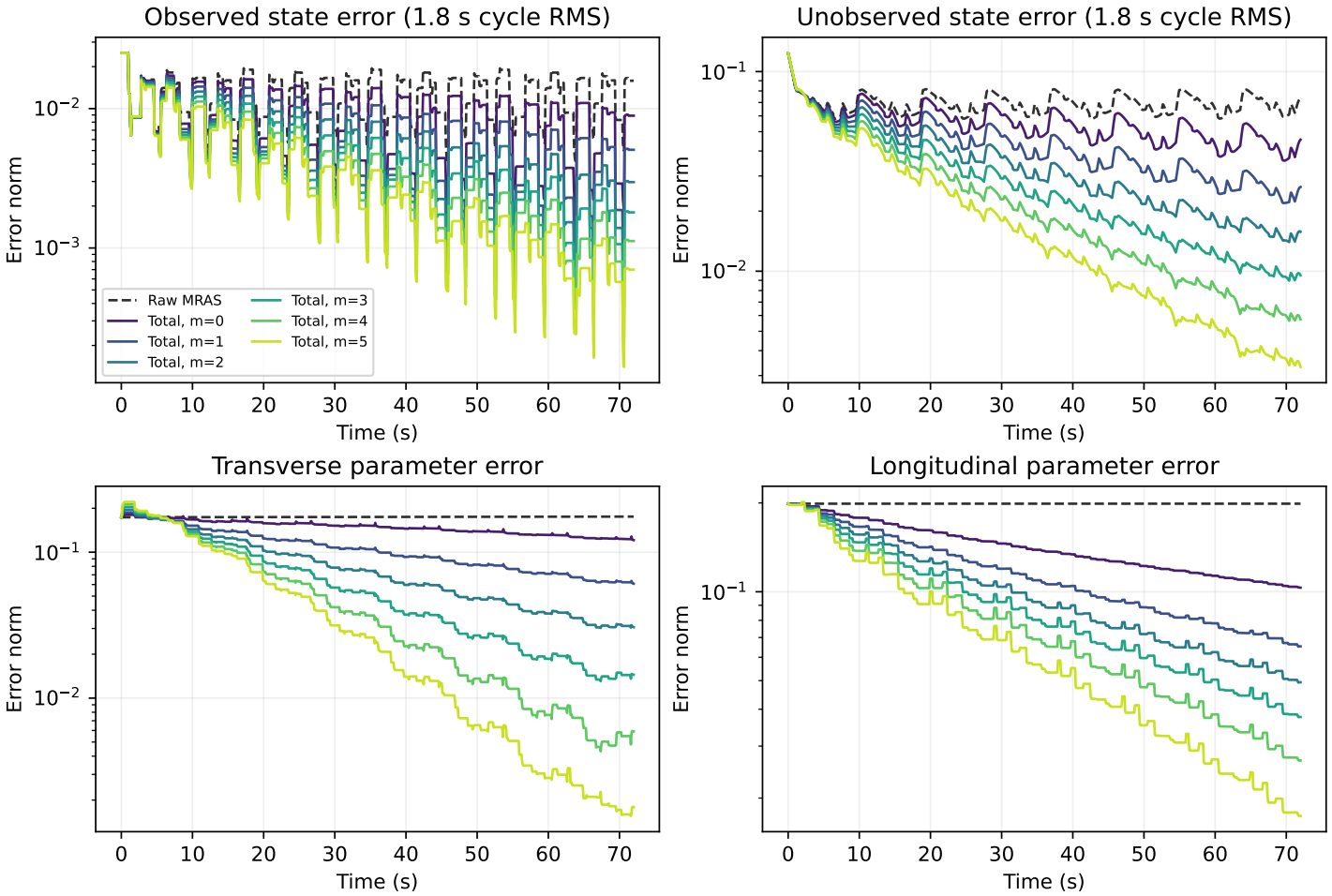
Numerical experiment



- 1D Bloch–Torrey model,
- $N = 9$, $\Delta t = 8$ ms,
- 12 unknown coefficients,
- 72 s record with a 36 s supercycle.

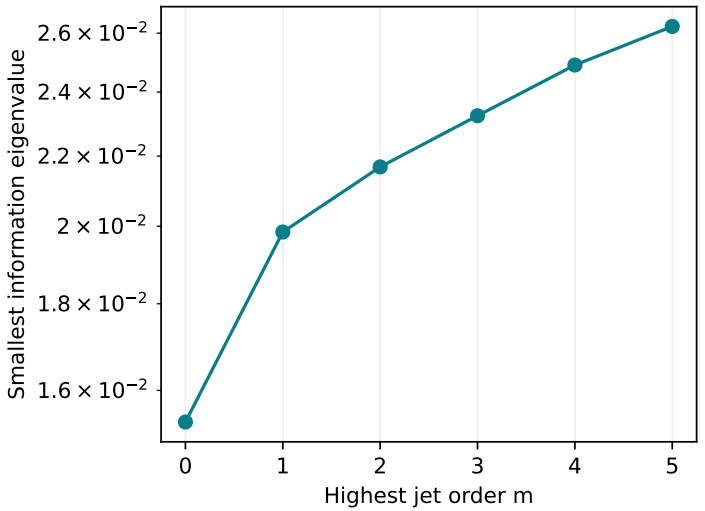
- exact noise-free jets, $m = 0, \ldots, 5$,
- the same sequence and initial error for every m,
- raw output alone corrects the state,

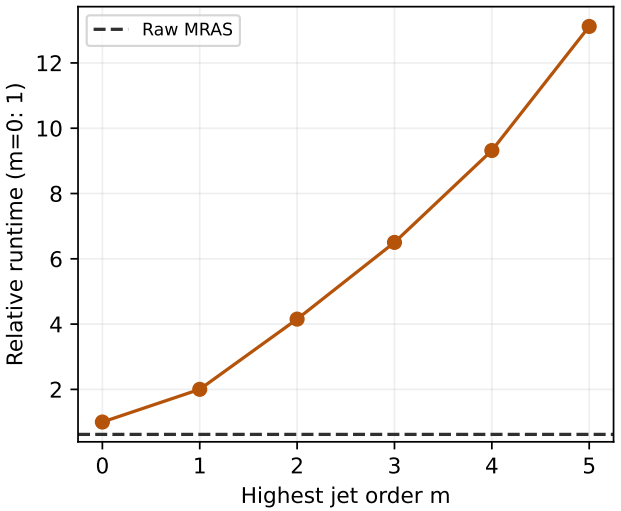
Total sensitivity already works at order zero



Raw MRAS leaves the longitudinal error unchanged. With $H_0 = R\widehat{S}$, the order-zero estimator uses delayed longitudinal information. Higher derivatives then accelerate convergence.

Information gain and computational cost





order	$\lambda_{\min}(\mathcal{I}_m)$	Floquet radius	$E(72)/E(0)$	relative runtime
$m = 0$	0.01533	0.80013	0.53529	1.0
$m = 2$	0.02168	0.52224	0.19414	4.2
$m = 5$	0.02624	0.28027	0.05734	13.1

More derivatives improve information and convergence, but the runtime rises quickly. Costs are normalised by the $m = 0$ run because absolute times depend on the hardware.

Limitations and next steps

Current limitations

- exact derivative channels without noise,
- small 1D discretisation and one sequence,
- continuum excitation and coupled PDE stability still open,
- desktop runtime benchmark still pending.

Next experiments

- causal derivative filters with a noise model,
- jet information in the state correction,
- more intricate numerical experiments.

Main next step

Add the sensitivity and jet residuals to the state correction, then analyse the resulting observer under noisy measurements.

Questions?

Thank you for your attention.